New chat

L9_S2_A.pdf
PDF

Generate a lecture scribe suitable for exam preparation from the provided lecture 9. While making this, reason step by step using definitions, assumptions, derivation, proof, and example solution exactly as given in the slides. Include the step by step reasoning so that when we revise for our exams it becomes easy. All explanations and reasoning must be from the provided ppt. do not introduce any new examples or theory apart from what is there in the ppt. Explain every definition and assumption explicitly. derive the formulas and solve the examples step by step using whatever is there in the slide. The tone should be formal and academically suitable for revision. try to include every example and question from the pdf

Absolutelyyyy buddy 💪📚
I prepared a **formal, exam-ready lecture scribe for Lecture 9 ONLY using what is present in your provided PPT/PDF** (no extra theory, no new examples).
This scribe includes **all definitions, assumptions, formulas, derivations, and examples exactly as in the slides**, with **step-by-step reasoning** for revision.

Source: Lecture 9 PDF

CSE400 — Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

---

1. Outline of Lecture (from slides)

This lecture covers **Types of Continuous Random Variables**, specifically:

1. **Uniform Random Variable**
   - Definition (PDF and CDF)

- Graph
- Example
- Application examples

2. **Exponential Random Variable**
    - Definition (PDF and CDF)
    - Graph
    - Example

(Laplace and Gamma are mentioned in the title/outline but this particular provided PDF segment contains Uniform + Exponential content only.)

---

2. Uniform Random Variable

2.1 Definition (PDF)

A continuous random variable X is said to be **Uniform** over the interval $[a, b)$ if its PDF is:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x < b \\ 0, & \text{elsewhere} \end{cases}$$

Interpretation:

- All values between a and b are **equally likely**.
- The height of the PDF is constant: $\frac{1}{b-a}$.

2.2 Definition (CDF)

The CDF of a uniform random variable is:

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leqslant x < b \\ 1, & x \geqslant b \end{cases}$$

Interpretation:

- For values before a, probability accumulated is 0.
- Between a and b, probability increases linearly.
- After b, probability becomes 1.

2.3 Graph (from slide figure)

From the diagram:

(a) PDF

- A flat horizontal line between a and b at height $\frac{1}{b-a}$.
- 0 everywhere else.

(b) CDF

- 0 until $x = a$.
- Then a straight line increasing from 0 to 1.
- Then constant at 1 after $x = b$.

---

3. Example 1 (Uniform Random Variable)

3.1 Problem Statement (given)

The phase of a sinusoid, Θ, is uniformly distributed over $[0, 2\pi)$.
So its PDF is:

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \leqslant \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

Key Uniform Property Used (from solution slide)

For a uniform RV on $[0, 2\pi)$:

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

(a) Find $\Pr(\Theta > \frac{3\pi}{4})$

Step-by-step:

- Total interval length = 2π

- Favorable region = $(\frac{3\pi}{4}, 2\pi)$

So:

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi}$$

Compute numerator:

$$2\pi - \frac{3\pi}{4} = \frac{8\pi}{4} - \frac{3\pi}{4} = \frac{5\pi}{4}$$

Thus:

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{\frac{5\pi}{4}}{2\pi} = \frac{5}{8}$$

✅ Final Answer:

$$\boxed{\frac{5}{8}}$$

(b) Find $\Pr(\Theta < \pi \mid \Theta > \frac{3\pi}{4})$

Step-by-step:

Let

- $A = \{\Theta < \pi\}$
- $B = \{\Theta > \frac{3\pi}{4}\}$

We use conditional probability formula:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

From part (a):

$$\Pr(B) = \Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{5}{8}$$

Now compute intersection:

$$A \cap B = \{\frac{3\pi}{4} < \Theta < \pi\}$$

Using uniform interval property:

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi}$$

Compute numerator:

$$\pi - \frac{3\pi}{4} = \frac{4\pi}{4} - \frac{3\pi}{4} = \frac{\pi}{4}$$

So:

$$\Pr(A \cap B) = \frac{\frac{\pi}{4}}{2\pi} = \frac{1}{8}$$

Now substitute:

$$\Pr(A \mid B) = \frac{\frac{1}{8}}{\frac{5}{8}} = \frac{1}{5}$$

✅ Final Answer:

$$\boxed{\frac{1}{5}}$$

(c) Find $\Pr(\cos(\Theta) < \frac{1}{2})$

Step-by-step:

We use the key trig identity given in slide:

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Now, cosine is **less than** $1/2$ in the interval:

$$\frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

So:

$$\Pr(\cos(\Theta) < \frac{1}{2}) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi}$$

Compute numerator:

$$\frac{5\pi}{3} - \frac{\pi}{3} = \frac{4\pi}{3}$$

Thus:

$$\Pr(\cos(\Theta) < \frac{1}{2}) = \frac{\frac{4\pi}{3}}{2\pi} = \frac{4}{6} = \frac{2}{3}$$

✅ Final Answer:

$$\boxed{\frac{2}{3}}$$

---

4. Uniform Random Variable: Application Examples

The slides list these direct applications:

1. **Phase of a sinusoidal signal**, where all phase angles between 0 and 2π are equally likely.
2. **Random number generated by a computer between 0 and 1** for simulations.
3. **Arrival time of a user within a known time window**, assuming no time preference.

---

5. Exponential Random Variable

5.1 PDF Definition (given)

The exponential random variable has PDF (for any $b > 0$):

$$f_X(x) = \frac{1}{b} \exp(-\frac{x}{b})\, u(x)$$

Here $u(x)$ is the unit step function.
This implies the PDF is only valid for $x \geqslant 0$.

5.2 CDF Definition (given)

The exponential random variable has CDF:

$$F_X(x) = [1 - \exp(-\frac{x}{b})]\, u(x)$$

Again, $u(x)$ ensures the distribution is only for $x \geqslant 0$.

5.3 Graph Interpretation (from slides)

From the plotted graphs:

PDF

- Starts at $\frac{1}{b}$ when $x = 0$
- Decreases exponentially toward 0 as $x \to \infty$

CDF

- Starts at 0 when $x = 0$
- Increases toward 1 as $x \to \infty$
- For example shown: $b = 2$

---

6. Example 2 (Exponential Random Variable)

6.1 Problem Statement

Let X be an exponential random variable with PDF:

$$f_X(x) = e^{-x} u(x)$$

Identify parameter b

General exponential PDF from slide:

$$f_X(x) = \frac{1}{b} \exp(-\frac{x}{b})\, u(x)$$

Given:

$$f_X(x) = e^{-x}u(x)$$

So we match:

- $\frac{1}{b} = 1 \Rightarrow b = 1$
- $-\frac{x}{b} = -x \Rightarrow b = 1$

Thus:

$$b = 1$$

(a) Find $\Pr(3X < 5)$

Step-by-step:

$$3X < 5 \Rightarrow X < \frac{5}{3}$$

Now use the CDF formula from slides:

For $b = 1$:

$$F_X(x) = 1 - e^{-x}$$

Thus:

$$\Pr\left(X < \frac{5}{3}\right) = F_X\left(\frac{5}{3}\right)$$

So:

$$\Pr(3X < 5) = 1 - e^{-\frac{5}{3}}$$

✅ Final Answer:

$$\boxed{1 - e^{-5/3}}$$

(b) Generalize to find $\Pr(3X < y)$

Step-by-step:

$$3X < y \Rightarrow X < \frac{y}{3}$$

So:

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right) = F_X\left(\frac{y}{3}\right)$$

Using $F_X(x) = 1 - e^{-x}$:

$$\Pr(3X < y) = 1 - e^{-y/3}$$

✅ Final Answer:

$$\boxed{1 - e^{-y/3}}$$

---

7. Final Exam Revision Summary (Lecture 9)

Uniform RV

- PDF:

$$f_X(x) = \frac{1}{b-a}, \quad a \leqslant x < b$$

- CDF:

$$F_X(x) = \frac{x-a}{b-a}, \quad a \leqslant x < b$$

- Probability in interval:

$$\Pr(a < X < b) = \frac{b-a}{\text{total length}}$$

Exponential RV

- PDF:

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

- CDF:

$$F_X(x) = [1 - e^{-x/b}]\, u(x)$$

If you want, I can also:

✅ convert this into **clean LaTeX** for your exam notebook

✅ make a **1-page cheat sheet** from this lecture

Just say the word buddy 😤🔥